

ENTERPRISE API TECHNICAL TASK VALIDATION TEST PLAN

Remove Use of GetMPCollectionByStudySubject from SubjectVisitTemplate

1. Technical Task Summary & Scope

1.1 Technical Task Identification

Technical Task ID	
Title	Remove Use of GetMPCollectionByStudySubject from SubjectVisitTemplate
Related Ticket	
Severity	Medium — Performance Degradation (large result-set over-fetch)
Priority	High — Product/Feature Bucket: Infrastructure — Performance
Work Category	Enhancement (Tech Task Type: Application Code Change)
Fix Version	
Invasiveness Score	3 — Low (no schema migration required)
API Changes Required	NO — Backend stored procedure only; API contract unchanged
Migration Test Cases	No — No schema or data migration needed
Test Automation	Candidate identified for API regression automation suite

1.2 Technical Task Description

The stored procedure GetMPCollectionForStudySubject fetches an oversized result set encompassing all Measurement Point (MP) records for a given study subject. SubjectVisitTemplate (SVT) then filters this large collection in-memory by FormID and Measurement Point ID to read a single field: IsSkipped. This is a classic N+1 / over-fetch anti-pattern that introduces unnecessary database load, increased memory pressure, and degraded response times, especially for subjects with many visits or forms.

The fix replaces this broad stored procedure call with a targeted, parameterized query that fetches only the specific IsSkipped value for the requested FormID and Measurement Point ID combination — eliminating the over-fetch entirely.

1.3 Affected API Endpoint(s)

Method	HTTP Verb	Path / Endpoint	API Version	Notes
GetFormInfo (or equivalent SVT loader)	GET	/api/subjects/{subjectId}/visits/{visitId}/forms/{formId}	v1 / current (no version change)	Backend now uses targeted SP instead of GetMPCollectionForStudySubject
SubjectVisitTemplate loader	GET	/api/subjects/{subjectId}/visit-template	v1 / current	Primary consumer of IsSkipped field — must return identical contract

1.4 Environment(s) in Scope

Environment	Status	Fix Deployed?	Notes
Development	Primary validation target	Yes — v42.12	Initial fix validation and DB-level verification
Feature Branch	Full regression execution	Required before QA sign-off	All test cases executed here
Dev Master	Full regression execution	Required before QA sign-off	All test cases executed here
Staging	Pre-production smoke test	Confirm same as QA	Performance baseline capture if not available in QA
Production	Out of scope for active testing	Post-release monitoring only	Monitor DB query stats post-deploy

1.5 Impacted Services & Downstream Consumers

- SubjectVisitTemplate (SVT) UI — primary rendering consumer of IsSkipped field.
- Backend API layer — the method/controller that reads IsSkipped for SVT display.
- Any OpenAPI external client or integration script consuming subject visit/form endpoints.
- Background jobs or reporting modules that derive skip status from the same data path
- Database layer — the stored procedure GetMPCollectionForStudySubject is being retired in this code path; DB query plans will change.

2. Root Cause Analysis (QA Perspective)

2.1 Failure Characterization

Failure Type: Over-fetching / Inefficient Data Access Pattern

Affected Layer: Data Access Layer (Stored Procedure) → Business Logic (In-Memory Filter) → API Response

Manifestation: Slow SVT load times; increased DB CPU and memory for large subjects.

Regression Risk: Low (Invasiveness Score 3) — single SP call replaced with targeted query.

2.2 Likely Failure Points

Layer	Pre-Fix Behavior (Root Cause)	Post-Fix Expected Behavior
Stored Procedure	GetMPCollectionForStudySubject returns ALL MP records for the study subject — unbounded result set	New targeted SP/query returns only the row matching (FormID, MeasurementPointID) — 1-row response
Business Logic / Filter	In-memory LINQ/loop filters the full collection down to one record — CPU-wasteful, allocates large objects	No in-memory filtering needed; data arrives pre-filtered from DB
API Response / Contract	IsSkipped value is correct (functionality preserved) but at high cost	IsSkipped value identical — no semantic or schema change to the API contract
Performance	SVT load degrades linearly as subject visit/form count grows; DB I/O spikes	SVT load time stable or improved; DB I/O significantly reduced for subjects with many forms

2.3 Assumptions & Open Questions for Dev Team

#	Assumption / Question	Expected Answer / Action Required
A1	The new stored proc / parameterized query is indexed on (FormID, MeasurementPointID) for optimal retrieval.	Confirm index exists. If not, request Dev/DBA to add it before QA sign-off.
A2	No concurrency issues arise from the new targeted query under simultaneous SVT loads.	Verify with Dev that the new SP is read-only with no locking implications.
A3	The change is confined to SubjectVisitTemplate and does not touch other consumers of GetMPCollectionForStudySubject.	Dev to confirm search/grep across codebase that only SVT called the old SP in this pattern.
A4	NULL / missing FormID or MeasurementPointID is handled gracefully by the new query (no unhandled exceptions).	QA will test this in TC8. Dev to confirm the SP has NULL-guard logic.
A5	Pre-change performance baseline numbers are available from logs or APM tooling.	Request Dev/DevOps to share pre-change response time samples for SVT endpoint before QA starts.

3. Test Objectives

3.1 Primary Objective

Validate that the removal of GetMPCollectionForStudySubject from SubjectVisitTemplate does not alter the functional behavior of IsSkipped for any form/measurement point combination, and that the new targeted data access approach returns identical, correct results to the previous implementation.

3.2 Secondary Objectives

Objective Area	Detail
Regression Coverage	Confirm all SVT-adjacent functionality (dates, statuses, completion %, access controls) remains unaffected.
Contract Integrity	Verify that the API response schema (field names, types, HTTP status codes) is unchanged for all SVT endpoints. Zero breaking changes permitted.
Performance Validation	Measure and document SVT load times pre- vs. post-change. Confirm no degradation; target measurable improvement for large-data subjects.
Security Posture	Ensure the new targeted query does not inadvertently expose data across role boundaries or study boundaries (no over-privilege in new SP).
Automation Readiness	Identify at least one representative API test case (IsSkipped for skipped/non-skipped form) as a candidate for inclusion in the automated API regression suite.

4. Test Scenarios & Test Cases

4.1 Technical Task Reproduction — Confirm Original Defect Symptom

These cases establish the pre-fix baseline. Execute against a pre-fix build (if available in a parallel environment) to document the defect symptom (slow load, large DB result set) before validating the fix.

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
BR-01	Confirm oversized SP result set on large subject	GET	/api/subjects/{subjectId}/visit-template	Pre-fix build available; subject has 50+ visits/forms; DB profiler enabled	SubjectID with 50+ form MP records	1. Enable DB query profiler 2. Call SVT endpoint 3. Capture rows returned by GetMPCollectionForStudySubject	SP returns ALL MP rows for subject (not filtered); large row count visible in profiler	[TBD]	[]
BR-02	Verify slow SVT load time on pre-fix build	GET	/api/subjects/{subjectId}/visit-template	Pre-fix build; large-data subject (100+ forms)	High-cardinality SubjectID	1. Record timestamp before request 2. Execute GET SVT endpoint 3. Record timestamp after response 4. Calculate elapsed time	Response time > post-fix baseline (establishes regression floor)	[TBD]	[]

4.2 Fix Validation — Confirm Resolution

The core pass/fail gate for this ticket. All cases must pass before QA sign-off.

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
FV-01	Non-skipped form displays correctly in SVT after fix	GET	/api/subjects/{id}/visit-template	Fix deployed (v42.12); subject has visit with ≥1 non-skipped form	SubjectID with known non-skipped FormID + MPID	1. Authenticate as valid user 2. GET /api/subjects/{subjectId}/visit-template 3. Locate form in response 4. Assert IsSkipped field value	IsSkipped = false; form appears without skip indicator; HTTP 200; response body schema unchanged	[TBD]	[]
FV-02	Skipped form displays correctly in SVT after fix	GET	/api/subjects/{id}/visit-template	Fix deployed; subject has ≥1 form marked as skipped	SubjectID + FormID + MPID where IsSkipped = true	1. Authenticate 2. GET SVT endpoint 3. Locate skipped form in response 4. Assert IsSkipped = true 5. Validate skip indicator present in UI	IsSkipped = true; skip indicator shown; restricted actions remain restricted; HTTP 200	[TBD]	[]
FV-03	Transition: non-skipped → skipped reflected in SVT	GET/PUT	/api/forms/{formId}/skip + GET /visit-template	Form initially non-skipped	FormID in active (non-skipped) state	1. PUT /forms/{formId}/skip to mark as skipped 2. GET SVT endpoint 3. Assert IsSkipped = true for that form	IsSkipped transitions to true; no stale cache returned; consistent across UI and API	[TBD]	[]
FV-04	Transition: skipped → non-skipped reflected in SVT	PUT/GET	/api/forms/{formId}/unskip + GET /visit-template	Form currently marked as skipped	FormID with IsSkipped = true	1. PUT /forms/{formId}/unskip 2. GET SVT endpoint 3. Assert IsSkipped = false 4. Confirm no residual skip indicators	IsSkipped = false; form fully active; no mixed state across views	[TBD]	[]
FV-05	Multiple forms with mixed skip states — correct per-form mapping	GET	/api/subjects/{id}/visit-template	Visit contains ≥3 forms: some skipped, some not	SubjectID with mixed-state forms	1. GET SVT endpoint 2. For each form in response, assert IsSkipped matches known state 3. Verify no off-by-one / ID mapping errors	Each form's IsSkipped value is independently correct; no cross-contamination	[TBD]	[]
FV-06	New SP returns only requested	DB/GET	New targeted SP (via Dev/DBA access or APM)	DB profiler / query log enabled; known		1. Trigger SVT load for single form 2. Capture DB query log 3. Count	SP returns exactly 1 row for the (FormID,	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
	row (targeted fetch verified)			(FormID, MPID) pair		rows returned by new SP 4. Assert row count = 1	MPID) pair — not the full subject collection		

4.3 Boundary & Edge Cases

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
BE-01	Non-existent FormID — graceful handling	GET	/api/subjects/{id}/visit-template?formId={invalid}	Valid subjectId; non-existent formId (e.g., 999999)	formId=999999 (guaranteed non-existent)	1. Authenticate 2. GET endpoint with invalid formId 3. Assert HTTP status and error body	HTTP 404 or 400 with descriptive error message; no unhandled exception or 500	[TBD]	[]
BE-02	Non-existent MeasurementPointID — graceful handling	GET	/api/subjects/{id}/forms/{formId}	Valid formId; non-existent MPID	mpId=0 or mpId=null	1. Call endpoint with invalid MPID 2. Assert response does not throw unhandled exception 3. Verify error response matches pre-fix behavior	HTTP 404/400; error payload schema unchanged from pre-fix behavior	[TBD]	[]
BE-03	Subject with zero forms / zero visits	GET	/api/subjects/{id}/visit-template	SubjectID exists but has no visits or forms configured	SubjectID with empty visit template	1. GET SVT endpoint for empty subject 2. Assert response body 3. Assert no DB exception from new SP	HTTP 200; empty array or empty visit template returned; no server error	[TBD]	[]
BE-04	Subject with maximum volume of visits and forms	GET	/api/subjects/{id}/visit-template	Subject with 200+ visits and	Large-volume SubjectID (stress subject)	1. GET SVT endpoint 2. Verify all forms listed	All forms listed; IsSkipped	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
				1000+ form/MP combinations		correctly 3. Verify IsSkipped correct for sample of forms 4. Record response time	correct; response time within SLA (see Section 6); no timeout		
BE-05	FormID with special characters in URL path	GET	/api/subjects/{id}/visit-template	FormID containing URL-encodable characters if applicable	formId with dash, underscore, or numeric only variants	1. Construct URL with edge-case FormID 2. GET endpoint 3. Assert correct IsSkipped returned	HTTP 200; correct IsSkipped value; no routing error or SQL injection vector	[TBD]	[]

4.4 Authorization & Authentication

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
AA-01	Valid token — authorized role — happy path auth	GET	/api/subjects/{id}/visit-template	Valid JWT/session token for a user with SVT read access	Bearer token for authorized role	1. Include valid Authorization header 2. GET SVT endpoint 3. Assert HTTP 200 and data returned	HTTP 200; IsSkipped data returned; no auth challenge	[TBD]	[]
AA-02	Expired token — rejected with 401	GET	/api/subjects/{id}/visit-template	Expired JWT token (expired_at in past)	Expired Bearer token	1. Include expired Authorization header 2. GET SVT endpoint 3. Assert HTTP 401	HTTP 401 Unauthorized; no SVT data returned; error body matches existing auth error schema	[TBD]	[]
AA-03	Missing token — rejected with 401	GET	/api/subjects/{id}/visit-template	No Authorization header sent	No auth header	1. Send GET request with no Authorization header 2. Assert HTTP 401	HTTP 401; error message "Unauthorized" or equivalent; no data leak	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
AA-04	Wrong role — insufficient permissions — 403	GET	/api/subjects/{id}/visit-template	Valid token for a role without SVT read access (e.g., read-only reporting role)	Bearer token for restricted role	1. Include valid token for restricted role 2. GET SVT endpoint 3. Assert HTTP 403	HTTP 403 Forbidden; no SVT data returned; new targeted SP does not bypass RBAC check	[TBD]	[]
AA-05	Cross-study data boundary — no data leakage from new SP	GET	/api/subjects/{id}/visit-template	User authenticated for Study A; SubjectID belongs to Study B	Cross-study SubjectID	1. Authenticate for Study A 2. GET SVT for Study B subject 3. Assert no data returned from Study B	HTTP 403 or 404; new targeted SP does not return data outside the authorized study scope	[TBD]	[]

4.5 Error Handling — 4xx / 5xx Scenarios

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
EH-01	Invalid SubjectID format — 400 Bad Request	GET	/api/subjects/INVALID/visit-template	No precondition	subjectId = "INVALID" (non-numeric or GUID format violation)	1. GET endpoint with malformed subjectId 2. Assert HTTP status and body	HTTP 400; descriptive validation error; no 500 from new SP receiving unexpected input type	[TBD]	[]
EH-02	DB connection failure simulation — 503/500	GET	/api/subjects/{id}/visit-template	DB connection temporarily unavailable (simulate with DBA or env config)	Any valid SubjectID	1. Disable/throttle DB connection 2. GET SVT endpoint 3. Assert response code and error body	HTTP 503 or 500; graceful error message; no raw SP error details exposed in response	[TBD]	[]
EH-03	Error message schema validation	GET	/api/subjects/9999999/visit-template	9999999 guaranteed to not exist	Non-existent SubjectID	1. GET endpoint with non-existent SubjectID 2. Assert HTTP 404 3. Validate error body schema: { "error":	HTTP 404; error body adheres to existing error schema (no	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
						string, "code": int, "message": string }	new fields introduced, no fields removed)		

4.6 Contract Testing — Schema, Headers, Content-Type

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
CT-01	Response body schema unchanged — IsSkipped field present and typed	GET	/api/subjects/{id}/visit-template	Valid authenticated request	Known SubjectID with forms	1. GET SVT endpoint 2. Parse JSON response 3. Assert IsSkipped field exists in form object 4. Assert IsSkipped is boolean type 5. Diff schema against OpenAPI spec or pre-fix baseline	IsSkipped: boolean present; no fields added or removed; schema matches OpenAPI contract exactly	[TBD]	[]
CT-02	Content-Type header = application/json	GET	/api/subjects/{id}/visit-template	Valid request	Any valid SubjectID	1. GET endpoint 2. Assert Content-Type response header	Content-Type: application/json; charset=utf-8 (unchanged from pre-fix)	[TBD]	[]
CT-03	Required response headers present	GET	/api/subjects/{id}/visit-template	Valid request	Any valid SubjectID	1. GET endpoint 2. Assert presence of: X-Request-Id, Cache-Control, Strict-Transport-Security (or per your API standard)	All standard security and correlation headers present; no new headers added or removed	[TBD]	[]
CT-04	HTTP status codes unchanged for all standard paths	GET	/api/subjects/{id}/visit-template	Execute with valid, expired, and missing tokens	Multiple auth states	1. Valid auth → assert 200 2. Expired token → assert 401 3. Not found → assert 404 4. Forbidden role → assert 403	Identical HTTP status codes to pre-fix behavior; no status code regression introduced by SP change	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
CT-05	OpenAPI backward compatibility — external client	GET	/api/subjects/{id}/visit-template	Existing integration or API consumer available	Standard integration payload	1. Trigger typical integration flow 2. Assert no parsing errors 3. Assert all expected fields consumed successfully	Zero breaking changes; integration logs show no new errors or contract violations	[TBD]	[]

4.7 Regression Testing

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
RG-01	SVT fields unrelated to IsSkipped are unchanged	GET	/api/subjects/{id}/visit-template	Known subject with full visit data	SubjectID with dates, statuses, completion %	1. GET SVT endpoint 2. Assert visit dates, form statuses, completion percentages, visit labels match pre-fix values	All non-IsSkipped fields unchanged; no collateral impact from SP change	[TBD]	[]
RG-02	Background jobs / reports using same data path unaffected	N/A	Internal job / report API endpoint	Background job using IsSkipped-derived data has run post-fix	Job output / report result	1. Trigger or wait for background job to run 2. Inspect output / logs 3. Compare to pre-fix job output for same data set	Job output identical to pre-fix baseline; no new exceptions in logs; no anomalies in report	[TBD]	[]
RG-03	User role permissions unchanged after SP swap	GET	/api/subjects/{id}/visit-template	Multiple user roles available: admin, coordinator, read-only	Tokens for each role type	1. For each role, GET SVT endpoint 2. Assert data visibility matches expected RBAC for that role	Access controls identical to pre-fix; no new data accessible or hidden due to SP change	[TBD]	[]
RG-04	Other SSR/reporting features on same DB remain unaffected	GET	/api/reports/subject-summary (or equivalent)	Related SSR endpoints available	Standard report request	1. Execute related SSR report 2. Assert output unchanged 3. Check for no new DB errors in logs	SSR reports unaffected; old SP retirement in SVT path does not	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
							cascade to other consumers		

4.8 Performance Testing

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
PF-01	SVT load time — moderate subject (baseline vs post-fix)	GET	/api/subjects/{id}/visit-template	Moderate subject: 10-20 visits, 30-50 forms; timing tool active	Moderate-volume SubjectID; pre-fix timing baseline from logs	1. GET endpoint 5x and record average response time 2. Compare to pre-fix baseline 3. Document method used (browser devtools / APM / k6)	Post-fix average ≤ pre-fix average; ideally measurable improvement (> 10% faster)	[TBD]	[]
PF-02	SVT load time — large subject (worst-case performance)	GET	/api/subjects/{id}/visit-template	Large subject: 100+ visits, 500+ forms; DB profiler enabled	High-volume SubjectID	1. GET endpoint 5x 2. Record average and p95 response time 3. Capture DB query row count and duration from profiler 4. Compare to pre-fix baseline	Significant improvement expected (large over-fetch eliminated); p95 < 2000ms; DB row count = 1 per form (not full collection)	[TBD]	[]
PF-03	Document performance measurement method	N/A	N/A	Pre-fix baseline data available or documented as unavailable	Performance metrics from pre- and post-fix	1. Choose and document measurement method (browser devtools / APM / k6 / DB profiler) 2. Record pre-fix and post-fix numbers 3. Attach or link results in JIRA ticket	Measurement method documented in ticket Q&A; performance data attached; result traceable per ticket requirement	[TBD]	[]
PF-04	DB resource utilization —	N/A	DB monitoring / APM	DB performance monitoring available	Large-volume SubjectID	1. Enable DB monitoring 2. Trigger SVT endpoint for large	DB CPU and I/O reduced vs pre-fix; no abnormal	[TBD]	[]

TC ID	Scenario	Method	Endpoint	Preconditions	Test Data	Steps	Expected Result	Actual Result	Pass/Fail
	CPU/memory during SVT load					subject 3. Capture CPU%, memory, I/O, query duration 4. Compare to pre-fix stats if available	resource spikes; query duration acceptable		

5. Test Data Requirements

5.1 Input Payloads

Category	Payload Type	Specification
Happy Path	GET with valid IDs	subjectId: {valid GUID or int}; visitId: {valid}; formId: {valid non-skipped form}; Authorization: Bearer {valid_token}
Happy Path	GET for skipped form	Same as above but formId maps to a form where IsSkipped = true in DB
Negative	Non-existent FormID	formId=999999 — guaranteed non-existent; expect 404
Negative	Invalid SubjectID type	subjectId="INVALID_STRING" or subjectId=0 — expect 400
Negative	Expired token	Authorization: Bearer {expired_jwt}; expect 401
Boundary	Max-volume subject	SubjectID with maximum configured visits and forms (200+ visits, 1000+ forms) — performance stress
Boundary	Zero-data subject	SubjectID with no visits or forms configured — empty state handling
Boundary	MPID = NULL	Measurement Point ID not provided or null — test SP NULL guard

5.2 Required Environment Variables & Secrets

- API_BASE_URL — Base URL for the target environment (e.g., <https://qa.medrio.com/api>).
- AUTH_TOKEN_VALID — Valid JWT or session token for a user with SVT read access.
- AUTH_TOKEN_EXPIRED — A pre-generated expired token for AA-02 / AA-03 tests.
- AUTH_TOKEN_RESTRICTED_ROLE — Token for a role without SVT access (for AA-04).
- SUBJECT_ID_MODERATE — SubjectID with 10–50 forms for standard functional tests.
- SUBJECT_ID_LARGE — SubjectID with 200+ forms for boundary / performance tests.
- SUBJECT_ID_EMPTY — SubjectID with zero visits/forms for empty-state tests.

- SUBJECT_ID_CROSS_STUDY — SubjectID belonging to a different study for AA-05.
- DB_PROFILER_ACCESS — DBA or APM access credentials for DB-level query verification (FV-06, PF-02, PF-04).

5.3 Dependent Data Setup

Dataset	Setup Method	Used By
Study A — Simple (1-2 subjects, mixed skip)	DB seed script or manual creation via admin UI; mark one form as skipped, one as active	FV-01, FV-02, TC baseline
Study B — Edge cases (state transitions)	Create subject with forms that have changed skip state; document pre/post state in test notes	FV-03, FV-04, BE-03
Study C — Large volume	DB seed with 200+ visits and 1000+ form/MP combinations; mix of skipped and active	BE-04, PF-01, PF-02
Mock / Stub for SP failure	WireMock or Mockoon stub to simulate DB unavailability for EH-02	EH-02
Pre-fix performance baseline	Request from Dev/DevOps: APM logs or query duration stats from build prior to v42.12	BR-01, BR-02, PF-01, PF-02, PF-03

6. API Request & Response Validation Checklist

#	Validation Check	Notes / SLA
1	HTTP 200 returned for all valid authenticated GET requests to SVT endpoint	Validate on every functional test case execution
2	HTTP 401 returned for expired or missing Authorization token	AA-02, AA-03 must confirm unchanged 401 behavior
3	HTTP 403 returned for authenticated but insufficiently privileged role	AA-04; RBAC must not be bypassed by new SP
4	HTTP 404 returned for non-existent SubjectID or FormID	EH-03, BE-01; error body schema must be unchanged
5	HTTP 400 returned for malformed / type-invalid path parameters	EH-01; new SP must not receive raw invalid input without guard
6	Response body JSON schema matches pre-fix OpenAPI contract exactly	CT-01; IsSkipped: boolean present; no fields added/removed
7	IsSkipped field type is boolean (not string "true"/"false" or int 0/1)	Type coercion change would break API consumers
8	Content-Type: application/json; charset=utf-8 header present	CT-02
9	Required security headers present: X-Request-Id, Strict-Transport-Security, Cache-Control	CT-03; verify against your API security standard
10	Response time SLA ≤ 2000ms (p95) for moderate subjects under normal load	PF-01; record and compare against pre-fix baseline
11	Response time SLA ≤ 5000ms (p95) for maximum-volume subjects	PF-02; performance improvement expected; no regressions

#	Validation Check	Notes / SLA
12	DB SP returns exactly 1 row per (FormID, MeasurementPointID) request — not full collection	FV-06; core fix verification at DB layer
13	Idempotency: repeated GET calls return identical IsSkipped value (no state mutation on read)	GET is read-only; assert 3 consecutive calls return same result
14	No cross-study data leakage from new targeted SP (AA-05)	Critical security check; SP must be scoped to authorized study
15	Error response body does not expose raw SP name, SQL details, or stack trace	EH-02; security hardening check on error responses
16	Pagination / filtering: if SVT supports pagination, assert skip states are consistent across pages	Verify no IsSkipped inconsistency introduced by paging boundary

7. Entry & Exit Criteria

7.1 Entry Criteria

All the following must be satisfied before QA begins test execution:

#	Criterion	Owner
1	Fix for EDC-38555 is deployed to the QA/Test environment at version 42.12 or higher.	Dev/DevOps
2	Test data (Study A, B, C datasets) is seeded and verified in the target environment.	QA Engineer
3	Valid authentication tokens for all required roles (authorized, expired, restricted) are available.	QA Engineer / DevOps
4	DBA or APM access for DB-level query verification (FV-06, PF-02) is confirmed.	DBA / QA Lead
5	Pre-fix performance baseline (response times, DB query stats) is shared by Dev/DevOps.	Dev / DevOps
6	OpenAPI specification (or equivalent contract document) for SVT endpoints is available for schema diff.	Dev / Architect
7	This Test Plan has been reviewed and approved by QA Lead and Development Lead.	QA Lead / Dev Lead

7.2 Exit Criteria

Testing is complete and the ticket may be moved to Ready for Release when all of the following are met:

#	Criterion	Verification Method
1	All Fix Validation test cases (FV-01 through FV-06) pass with no failures.	Test execution report
2	All Contract Testing cases (CT-01 through CT-05) pass; API schema unchanged confirmed.	Pact / schema diff report

#	Criterion	Verification Method
3	All Authorization cases (AA-01 through AA-05) pass; no RBAC or data leakage defects.	Test execution report
4	All Regression cases (RG-01 through RG-04) pass; no collateral damage observed.	Test execution report
5	Performance tests (PF-01, PF-02) show no regression vs pre-fix baseline; results documented.	k6 report + ticket Q&A update
6	Performance measurement method and results are attached or linked in the JIRA ticket Q&A.	JIRA ticket updated by QA
7	No Critical or High severity defects remain open against EDC-38555.	JIRA defect triage
8	No new defects introduced by this fix in adjacent SVT functionality or related SSR features.	Regression test report + log review
9	Test Automation candidate (AT-01) is documented in ticket Test Automation Summary field.	JIRA ticket updated by QA

8. Risk & Mitigation

#	Risk	Likelihood	Impact	Mitigation	Owner
1	New targeted SP returns incorrect IsSkipped for edge-case (FormID, MPID) combinations — silent functional regression	Medium	Users see incorrect skip status in SVT; downstream reporting may reflect wrong data; clinical data integrity risk in EDC context.	FV-05 and BE-01–BE-05 cover boundary and mixed-state cases. DB-level row count assertion (FV-06) confirms targeted fetch. Baseline comparison (TC7 equivalent) for sampled forms.	QA Engineer
2	Pre-fix performance baseline unavailable — cannot prove improvement or detect regression	High	PF-01/PF-02 exit criteria cannot be met; performance objective of the ticket is unverifiable.	Escalate baseline request to Dev/DevOps as entry criterion (EC-05). If unavailable, capture detailed post-fix baseline and document this as the new reference for future comparisons. PF-03 documents the chosen measurement method regardless.	QA Lead / DevOps
3	Other codebase consumers of GetMPCollectionForStudySubject not identified — untested regression path	Low	Another feature using the same SP may behave differently if the SP is modified or removed; could cause production defects in unreported areas.	Require Dev to provide a codebase grep/search result confirming SVT is the only consumer affected by this change (assumption A3). RG-04 spot-checks related SSR features.	Dev + QA Lead
4	New targeted SP missing index on (FormID, MeasurementPointID) — performance improvement not realized or introduces table scan	Medium	Without proper indexing, the targeted SP may be slower than expected or cause table scans under load, negating the performance objective.	Assumption A1 — QA to explicitly confirm index existence with DBA before PF tests. PF-02 with DB profiler will surface execution plan issues.	DBA + Dev

#	Risk	Likelihood	Impact	Mitigation	Owner
5	Test environment instability delays execution — shared env contention or missing test data	Medium	Test delays, false failures, and unreliable performance measurements if environment is shared or unstable.	Request a dedicated or isolated test environment for performance runs. Ensure test data setup (EC-02) is scripted and repeatable. Coordinate with other QA teams for environment window.	QA Lead / DevOps

Appendix: Q&A

The following Q&A is pre-populated based on ticket metadata. QA and Dev should update answers upon completion of testing.

#	Question	Answer
1	Does this change alter any OpenAPI contract or field names?	No. Per ticket: "Will this require API changes?: NO." Only the underlying stored procedure usage was changed; API contracts remain identical.
2	How was performance validated?	[TO BE FILLED] Compared SVT load times (and backend query duration where available) for subjects with high form/visit counts before and after the change using [chosen method]. Results documented in the ticket and attached as CI artifact.
3	Are there any known limitations or edge cases around IsSkipped now?	[TO BE FILLED] None observed so far. Skipped and non-skipped forms behave identically to baseline in UI and API for all tested scenarios.
4	Is the new SP indexed on (FormID, MeasurementPointID)?	[TO BE FILLED — confirm with DBA before performance tests]
5	Which other services call GetMPCollectionForStudySubject?	[TO BE FILLED by Dev — codebase grep results expected as entry criterion]
6	Is an automated test candidate identified?	[TO BE FILLED] Yes / No — IsSkipped API test for skipped and non-skipped forms is a candidate for the API regression suite. Rationale documented in Test Automation Summary field.